

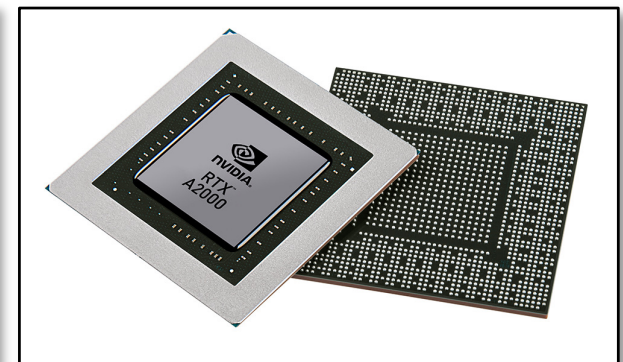
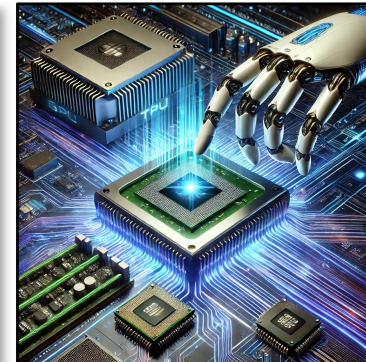
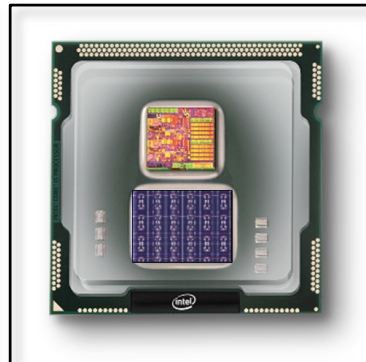
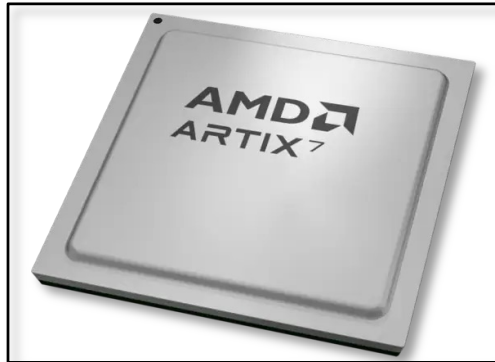
Christof Teuscher
ECE 410/510: Hardware for AI and ML, Spring 2026

Codefest #5

Portland State University
Department of Electrical and Computer Engineering (ECE)

www.teuscher-lab.com

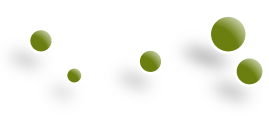
teuscher@pdx.edu



Quiz 1 format

- **Format:** Online oral examination on the *Shine* platform.
- **Why oral?** In the age of LLMs, it's perhaps the best/only way to assess your understanding. It's also great practice for your professional life.
- **Duration:** 20min
- **Number of questions:** 5 questions, 4 min per question
- **Grading:** S/NY (80% threshold). A quiz 1 repeat will consume 2 tokens
- The practice session will close at midnight today!

<https://teuscher-lab.ece.pdx.edu/shine>



Practice session grading

- Fully automated
- Tough grader!

Question Analysis

Based on 18 sessions

Table

■ Done ■ Silence ■ Time timeout

#	Question	Done	Silence	Timeout	Avg Time
1					49s
2					35s
3					34s

“Real” quiz 1 grading

- AI assisted with a lot of human oversight.
- What I’m looking for:
 - Do you understand the foundations? At what level?
 - Can you explain them to me?
 - Do you know what you are talking about?
 - I want to see your thinking.
- Different expectations for grad and undergrads.

Vibe Coding is not an excuse for low-quality work

A field guide to responsible AI-assisted development



ADDY OSMANI

APR 17, 2025



160



16



27

Share

“Move faster and break even more things.”



I Am Developer ✓

@iamdeveloper

vibe coding, where 2 engineers can now create the tech debt of at least 50 engineers



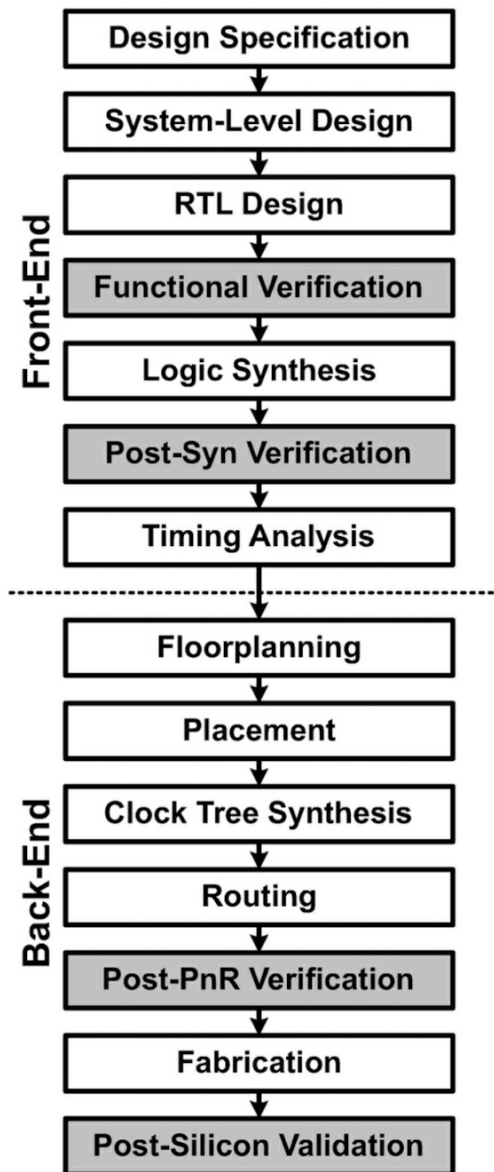
ASIC design flow

Mishra, Ashutosh; Cha, Jaekwang; Park, Hyunbin; Kim, Shiho. Artificial Intelligence and Hardware Accelerators.

Fig. 9 ASIC design flow

logic and architecture designs

physical design



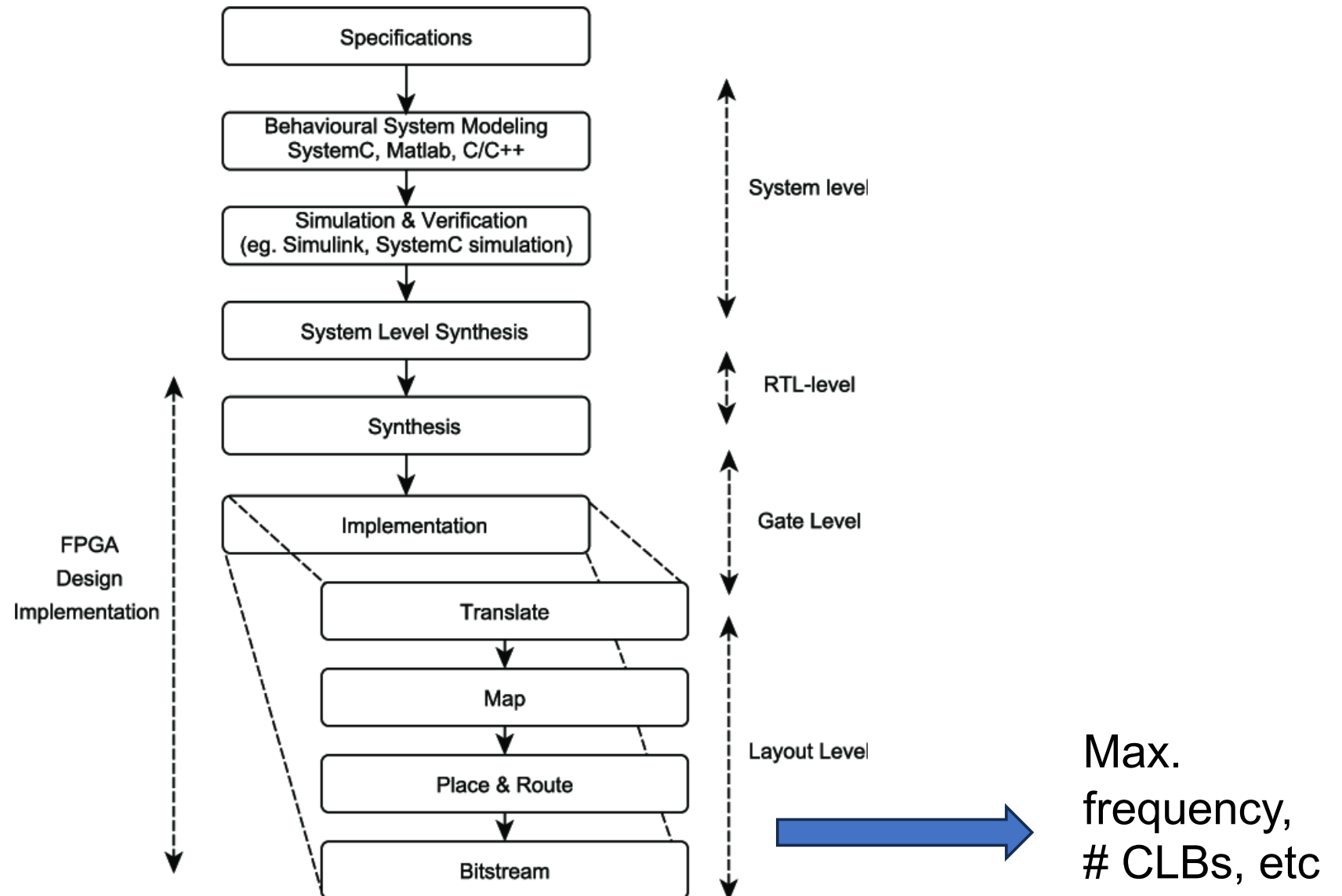
Python
Verilog/SystemVerilog/VHDL



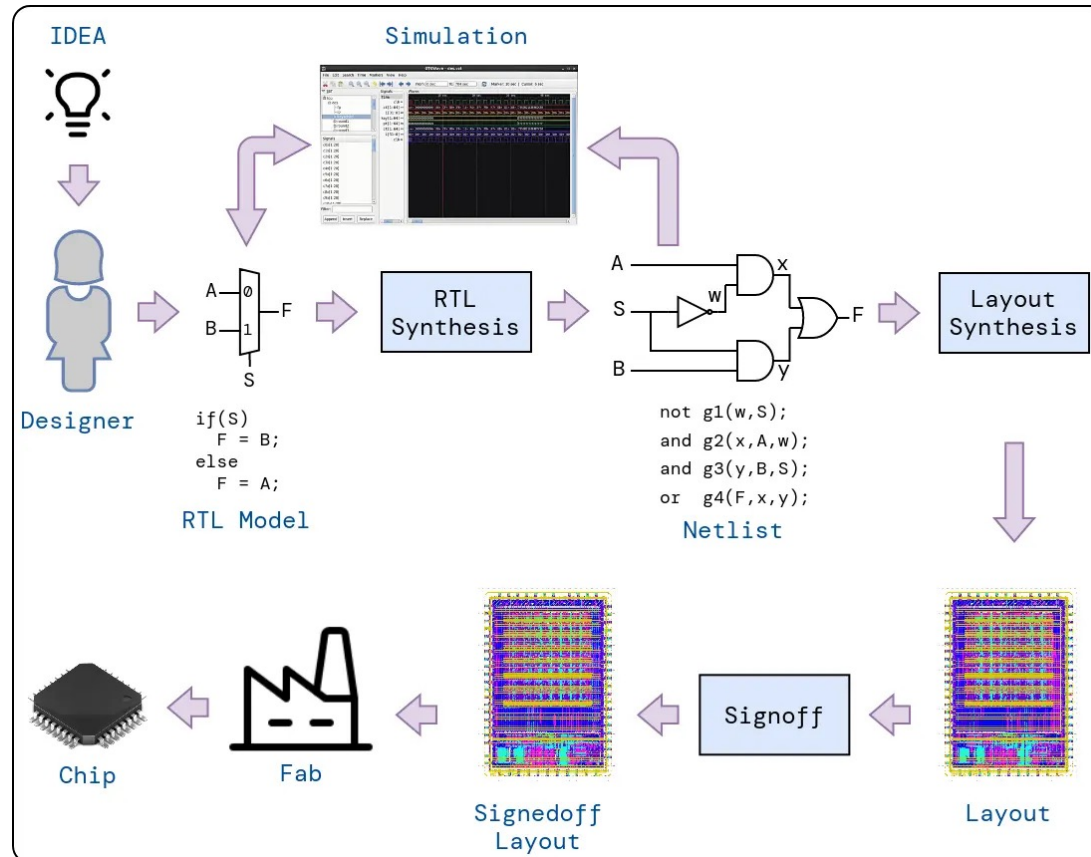
Max. frequency, #
transistors, etc.



FPGA design flow



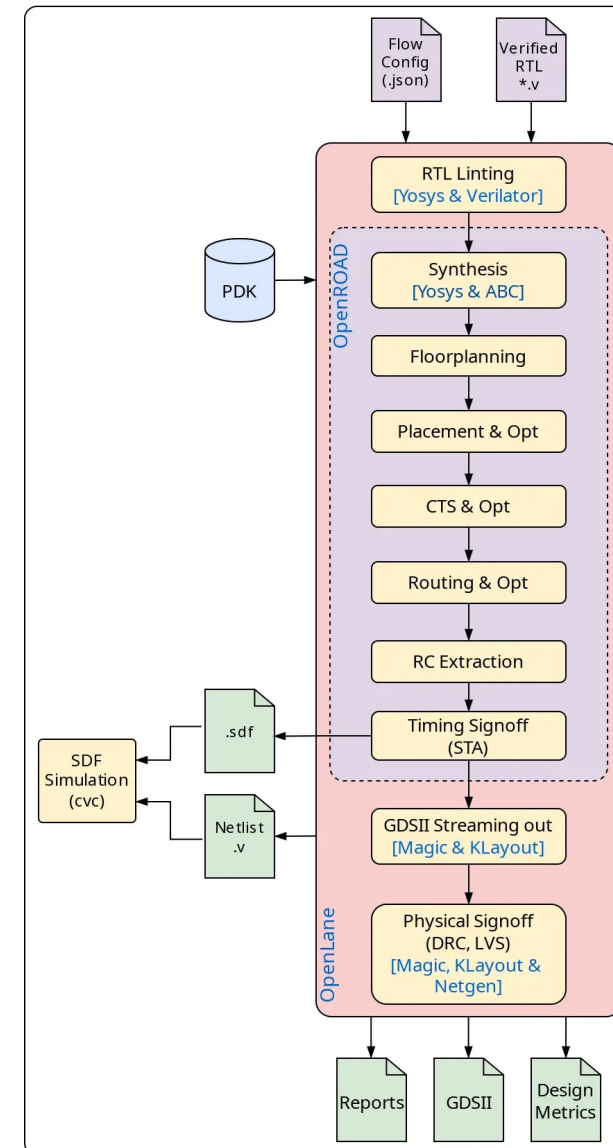
OpenLane 2 tools



<https://efabless.com/openlane>

<https://github.com/efabless/openlane2>

https://openlane2.readthedocs.io/en/latest/getting_started/newcomers/index.html



Getting started with OpenLane 2

To check out an example of an OpenLane 2-based flow right in your **browser**, try the Google Colab™ notebook at

<https://colab.research.google.com/github/efabless/openlane2/blob/main/notebook.ipynb>

To set up OpenLane 2 on your **computer**, check out the Getting Started guide at the following link: https://openlane2.readthedocs.io/en/latest/getting_started/index.html

Presentations (1)

- **Goals**
 - Share your idea
 - Share your progress
 - Hone your presentation skills
- **Schedule**
 - Presentation duration: 1 minute (strict). **Anybody who talks longer than 1 min is disqualified from extra tokens!**
 - Schedule linked on Canvas
- **Instructions**
 - Create a single (= 1 = ONE) slide on which you answer the following questions:
 - **What are you trying to do? I.e., what AI/ML algorithm/workload did you pick?**
 - **How have others implemented and/or accelerated this algorithm?**
 - **What are you doing differently/better/etc.?**
 - **What have you accomplished so far?**
 - **What will you do next and what remains to be done until you can declare success?**
- Save the slide in the shared Google folder. URL on Canvas
- A strong presentation earns 1 token.

Presentations (2)

Date	Week	Time	Name
Wed, April 22	4	4:55: PM	Eraganeni, Vamsidhar Reddy (Pronoun: He/Him)
Wed, April 22	4	4:57: PM	Kamarajugadda, Venkata Sriram (Pref: Venkata Sriram)
Wed, April 22	4	4:59: PM	Boss, David (Pronoun: He/Him)
Wed, April 22	4	5:01: PM	Hynes, Kay (Pref: Kay)
Wed, April 22	4	5:03: PM	Gilbert-Croysdale, Rebecca R. (Pronoun: She/Her)
Wed, April 22	4	5:05: PM	Charian, Sai Ganesh Reddy
Wed, April 29	5	4:55: PM	Vemula, Rajeshwar Reddy (Pronoun: he/him)
Wed, April 29	5	4:57: PM	Henrichs-Tarasenkov, Fyodor P. (Pref: Fedya, Pronoun: He/him)
Wed, April 29	5	4:59: PM	Herrera Moreno, Ivan (Pref: Ivan)
Wed, April 29	5	5:01: PM	Nguyen, Andrew (Pref: Bao)
Wed, April 29	5	5:03: PM	Chikani, Fatema Z. (Pronoun: She / Her)
Wed, April 29	5	5:05: PM	Munnangi, Pratibha (Pref: Pratibha, Pronoun: she/her)
Wed, May 6	6	4:55: PM	Ngo, Huy (Pref: Michael)
Wed, May 6	6	4:57: PM	Nguyen, Nhat M. (Pronoun: he/him)
Wed, May 6	6	4:59: PM	Naidu, Mahesh (Pref: Mahesh, Pronoun: He/Him)
Wed, May 6	6	5:01: PM	Rabiah, Yaqoub Y. (Pref: Yaqoub)
Wed, May 6	6	5:03: PM	Sabti, Saleh M. (Pref: Saleh, Pronoun: He/Him)
Wed, May 6	6	5:05: PM	Alali, Manaf A.
Wed, May 13	7	4:55: PM	Stanton, Andrew J. (Pref: Andrew, Pronoun: he/him)
Wed, May 13	7	4:57: PM	Bandla, Karthik
Wed, May 13	7	4:59: PM	Alrandi, Dawood A.
Wed, May 13	7	5:01: PM	Premarajan, Roshan B. (Pref: Roshan, Pronoun: hr/him)
Wed, May 13	7	5:03: PM	Vedantam, Venkata Krishna Kumar (Pronoun: His/Him)
Wed, May 13	7	5:05: PM	Esmail, Saleh A. (Pref: Sal, Pronoun: He/Him)

Wed, May 20	8	4:55: PM	Mills, Camden J. (Pref: Camden, Pronoun: He/Him)
Wed, May 20	8	4:57: PM	Day, Elliott F. (Pronoun: he/him)
Wed, May 20	8	4:59: PM	Mahajan, Ria (Pref: Ria)
Wed, May 20	8	5:01: PM	Khaddoori, Ibrahim A. (Pref: Ibrahim)
Wed, May 20	8	5:03: PM	AlQuzweni, Hussain A. (Pref: Hussain, Pronoun: he/him)
Wed, May 20	8	5:05: PM	Alotaibi, Mohammed M. (Pref: Mohammed)
Wed, May 27	9	4:55: PM	Wyse, Ian A. (Pronoun: he/him/his)
Wed, May 27	9	4:57: PM	Vanukuri, Sri Chandana Reddy
Wed, May 27	9	4:59: PM	Thakur, Amogh Vinod
Wed, May 27	9	5:01: PM	Kyatham, Vanisri (Pronoun: she/her)
Wed, May 27	9	5:03: PM	Vanamala, Naveen Babu (Pronoun: HE/HIM)
Wed, May 27	9	5:05: PM	Mallick, Devdutt S. (Pref: Devdutt, Pronoun: He/Him)
Wed, May 27	9	5:07: PM	Alshaiji, Mohammad A. (Pronoun: He/Him)
Wed, June 3	10	4:55: PM	Ramirez, Jose A. (Pronoun: he/him)
Wed, June 3	10	4:57: PM	Venkatesha Murthy, Veekshith (Pref: Veekshith, Pronoun: he, him)
Wed, June 3	10	4:59: PM	Handwerger, Adam L.
Wed, June 3	10	5:01: PM	Akula, Ranjith (Pronoun: he/him)
Wed, June 3	10	5:03: PM	Mickelson, Christian I. (Pref: Christian, Pronoun: He/Him)
Wed, June 3	10	5:05: PM	Shamszadeh, Navid (Pref: Navid, Pronoun: he/him)
Wed, June 3	10	5:07: PM	De Los Santos Razo, Jorge J. (Pronoun: He/Hims)